

immse Function Test Results

Test Case 1: Identical Image

"Matrix A:"

5. 5.

5. 5.

"Matrix B:"

5. 5.

5. 5.

"Difference Matrix:"

0. 0.

0. 0.

"MSE:" 0

Test Case 2: Small Difference

"Matrix A:"

5. 5.

5. 5.

"Matrix B:"

4. 6.

5. 5.

"Difference Matrix:"

1. -1.

0. 0.

"MSE:" 0.5

Test Case 3: Large Difference

"Matrix A:"

5. 5.

5. 5.

"Matrix B:"

0. 0.

0. 0.

"Difference Matrix:"

5. 5.

5. 5.

"MSE:" 25

Test Case 4: Float Input

"Matrix A:"

0.0863923 0.5841638 0.1793804 0.7072563

0.9672264 0.4442826 0.3433212 0.3831949

0.9007371 0.2022831 0.2862963 0.1268116

0.1160365 0.5732977 0.7422512 0.9768622

"Matrix B:"

0.1276622 0.1993201 0.8161063 0.5992154

0.7272676 0.6815347 0.9376651 0.0797154

0.6491194 0.3170464 0.9917201 0.4055675

0.9835055 0.9829524 0.8391549 0.6235772

"Difference Matrix:"

0.0875457 -0.5375949 0.7726379 -0.38883

0.007 -0.1262625 -0.4956071 0.5845436

0.5742582 0.2031317 -0.140077 0.0713961

-0.7601492 0.025134 0.6704808 -0.2095764

"MSE:" 0.1770277

Test Case 5: Negative Values

"Matrix A:"

-1. -2.

3. 4.

"Matrix B:"

1. 2.

-3. -4.

"Difference Matrix:"

-2. -4.

6. 8.

"MSE:" 30

Test Case 6: Size Mismatch

Output: "Error handled correctly"

Test Case 7: Zero Images

"Matrix A:"

0. 0. 0.

0. 0. 0.

0. 0. 0.

"Matrix B:"

0. 0. 0.

0. 0. 0.

0. 0. 0.

"Difference Matrix:"

0. 0. 0.

0. 0. 0.

0. 0. 0.

"MSE:" 0

Test Case 8: One Zero Image, One Non-Zero Image

"Matrix A:"

0. 0. 0.

0. 0. 0.

0. 0. 0.

"Matrix B:"

10. 10. 10.

10. 10. 10.

10. 10. 10.

"Difference Matrix:"

-10. -10. -10.

-10. -10. -10.

-10. -10. -10.

"MSE:" 100

Test Case 9: Large Values

"Matrix A:"

1000. 2000.

3000. 4000.

"Matrix B:"

1100. 1900.

3100. 3900.

"Difference Matrix:"

-100. 100.

-100. 100.

"MSE:" 10000

Test Case 10: Single Element

"Matrix A:"

5.

"Matrix B:"

10.

"Difference Matrix:"

-5.

"MSE:" 25

Test Case 11: Floating Precision Check

"Matrix A:"

0.1 0.2

0.3 0.4

"Matrix B:"

0.1 0.25

0.35 0.45

"Difference Matrix:"

0. -0.05

-0.05 -0.05

"MSE:" 0.0018750

Test Case 12: Large Random Matrix

"Matrix A Size:"

50. 50.

"Matrix B Size:"

50. 50.

"MSE:" 0.1681994

SUMMARY TABLE

TC	Description	Expected Output	Obtained Output
1	Identical Image	0	0
2	Small Difference	0.5	0.5
3	Large Difference	25	25
4	Floating-Point Input	Positive value	0.1770277
5	Negative Values	30	30
6	Size Mismatch	Error: Input images must have same dimensions	Error handled correctly
7	Zero Images	0	0
8	One Zero, One Non-zero	100	100
9	Large Values	10000	10000
10	Single Element	25	25
11	Floating Precision Check	0.001875	0.001875
12	Large Random Matrix	Positive value	0.1681994